Agent-Navigable Scientific Computing

Paper 1 • Complete research, engineering, experiment and publication blueprint

Working manuscript: “Scientific Capability Contracts for Reliable AI-Agent Discovery and Execution of Computational Workflows”

Status: preregistration-ready planning document, not a completed systematic review, validated technology, or manuscript reporting results. All quantitative targets are proposed designs, not observations.

Prepared: 25 October 2026 | Intended initial domain: neuroimaging | Reusable public tool: SciCap

Executive decision

The practical opportunity is not another command-line wrapper or an MCP server. Existing workflow engines and registries already provide typed interfaces, containers, execution, and workflow sharing. The testable contribution is a domain-aware, machine-readable scientific capability contract that improves AI agents’ tool discovery, configuration, and task execution while preserving reproducibility and meaningful failure reporting. The primary scientific claim must be demonstrated experimentally against strong existing interfaces, with ablations and held-out tasks.

Track	Primary deliverable	Go/no-go standard
Research	Controlled methods study	Meaningful improvement over matched standard tool interfaces; credible external validity
Software	SciCap Python package + CLI + MCP adapter	Independently installable and useful without an LLM
Community	Open catalog, docs, examples and feedback	External users can publish/inspect contracts; documented issues
Publication	Methods paper or software paper	Journal choice decided by demonstrated novelty and breadth

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1. Concept and precise contribution

1.1 Motivation

Scientific tools are heterogeneous across domains, software stacks and compute settings. AI agents can interpret documentation and invoke APIs, but tool descriptions frequently omit scientific meaning: target modality, measurement semantics, coordinate reference, preconditions, output interpretation, validation obligations and reasons to abstain. Existing engines and schemas often encode syntax and dependencies well; the proposed work investigates whether additional machine-readable scientific metadata measurably improves agent decision-making.

1.2 Problem statement

Given user intent U, available capability set C, input dataset D and execution policy P, an agent must select an appropriate capability c in C, construct valid arguments a, identify missing preconditions, request permitted execution, and produce a traceable result. Performance is not satisfied by a plausible answer or successful process exit alone. The framework should distinguish correct selection, runnable configuration, scientifically compatible input, successful execution and verifiable result.

1.3 Intended contribution (contingent on results)

- A versioned scientific capability contract extending existing tool descriptors with explicit domain semantics and machine-checkable preconditions.
- A registry and adapter architecture that can expose a single capability to human interfaces, CLIs and agent protocols without duplicating its definition.
- An evaluation that isolates the incremental value of scientific semantic metadata over conventional descriptions, typed interfaces and documentation.
- An open neuroimaging reference implementation with accessible fixtures, provenance output and published negative cases.

1.4 Explicit non-claims

- Do not claim invention of workflow DAGs, typed software interfaces, containers or reproducibility: Nipype, Boutiques, CWL, BIDS Apps and other projects predate this work.
- Do not claim that a contract guarantees scientific validity: some checks require image inspection, expert judgment, statistical design or downstream QC.
- Do not claim general cross-domain applicability without testing on a second domain and independent tools.
- Do not claim that improvements in a single agent/model demonstrate architecture independence.

1.5 Primary outcomes for public users

SciCap allows developers to inspect and publish scientific capabilities; researchers to query requirements, test compatibility and generate valid invocation templates; and AI agents to discover tools through standard adapters. The base registry, validation and documentation generation must remain functional when no model API is configured.

2. Prior art and literature synthesis

2.1 Related-work landscape

Work / standard	Established contribution	What SciCap must add or test

Nipype (2011) [1]	Interfaces to neuroimaging tools, typed inputs, workflow DAGs, execution plugins	Agent-oriented semantic choice and measured reliability beyond those interfaces
BIDS (2016) [2] / BIDS Apps (2017) [3]	Dataset organization, standardized app invocation/reproducibility	Scientific capability semantics and task-intent mapping, reusing BIDS
Boutiques (2018) [4]	Portable JSON descriptors of CLI tools, validators and container links	Distinct semantic fields, not duplication of descriptor mechanics
fMRIPrep (2019) [5]	Robust, tested, data-adaptive neuroimaging processing	Use as exemplar of mature pipelines; do not rebuild preprocessing
CWL [6] / workflow engines	Interoperable workflow and tool execution descriptions	Compare against native tool/workflow representations
RO-Crate / W3C PROV [7,8]	Research-object metadata and provenance relationships	Produce compatible provenance rather than proprietary logs
MCP [9] / OpenAPI [10]	Tool discovery and machine-readable call schemas	Scientific domain semantics and preflight limitations; use MCP as transport
Balis et al. (2026) [11]	Agentic research-intent interpretation + deterministic workflow generation and skills	Differentiate from skill prompts; show gains from inspectable typed contracts
Paper2Agent (2026) [12]	Converts papers/code into MCP tools and validates executable paper agents	Not a paper-agent conversion; focus on scientific tool catalog and controlled metadata effects

2.2 Literature-derived lessons

- The strongest baseline is not a raw LLM with shell access alone: it includes established typed descriptors (e.g., Boutiques/CWL/Nipype) and MCP-exposed schema tools.
- Scientific metadata should be an explicit intervention: if all conditions receive different wording or examples, any improvement may be due to prompt engineering rather than representation.
- Provenance, containerization and input validation are baseline engineering requirements, not novel results in isolation.
- Agentic science papers published in 2026 raise the burden of evidence: use multiple tasks, realistic failures, ablations, independent executions and transparent costs.

2.3 Claims requiring further verification before submission

This is a targeted narrative scan, not yet a systematic literature review. Before fixing novelty, search workflow type systems, semantic web services, OWL-S/SAWSDL, scientific ontology-driven planning, AI tool-selection benchmarks, agent safety/permission models, Nipype interface traits, Boutiques descriptors, BIDS Apps metadata and MCP ecosystems. Record inclusion decisions and check 2026–2027 papers close to submission.

3. Research questions and hypotheses

ID	Research question	Testable hypothesis
RQ1	Does enriched scientific metadata improve capability selection?	H1: semantic-contract condition improves held-out task-selection accuracy versus matched schema tools.
RQ2	Does it improve correct argument configuration and abstention?	H2: fewer invalid configurations and better calibrated deferral when preconditions are absent.
RQ3	Which contract fields cause improvement?	H3: ablations of modality, scientific type and precondition fields reduce performance.
RQ4	Does it generalize?	H4: benefit persists on unseen tools, datasets and at least one other research

		domain.
RQ5	Does it add efficiency costs?	Exploratory: quantify latency, token usage, authoring effort and compute overhead.

Declare one primary endpoint before confirmatory evaluation, e.g., rate of scientifically correct completed tasks on a held-out set. Hypotheses above are proposals; do not report as findings.

3.1 Scope boundary with Paper 2

Paper 1 evaluates scientific capability discovery and correct invocation. Paper 2 separately investigates formal compatibility verification and automatic composition of multi-step workflows. Paper 1 may implement limited precondition checks, but it should not claim a new formal verification method or use Paper 2's principal experiments as its own.

4. SciCap public software: functional specification

4.1 Personas and use cases

Persona	User job	Success criterion
Scientific software developer	Publish a tool with a validated machine-readable contract	Contract passes schema tests and yields usable API/CLI docs
Neuroimaging researcher	Find an appropriate processing operation for a dataset	Correct requirements shown with warnings and provenance expectations
AI agent developer	Discover and invoke approved scientific functions	Structured arguments, permission boundary, machine-readable errors
Workflow maintainer	Reuse capability definitions in pipelines	No divergence between CLI/MCP/docs representation

4.2 Minimum product (MVP)

- Capability schema v0.1 with stable IDs, version, purpose, semantic inputs/outputs, preconditions, resources, execution references and validation hooks.
- Offline CLI: list, search, inspect, validate, explain, export, and dry-run.
- Python SDK and Pydantic models with JSON Schema publication.
- Read-only catalog web interface or documentation site; no patient upload required.
- MCP adapter providing list/detail/validated-invoke interfaces, with user approval at execution boundary.
- Five initial capabilities: BIDS validation, anatomical segmentation, volume extraction, image registration and simple regional-statistics operation.
- Machine-readable errors and provenance bundle; examples with permitted public data.

4.3 Contract schema: proposed fields

Group	Fields	Reason
Identity	id, version, maintainers, license, source, citation	Discovery, maintenance, citation
Scientific meaning	task taxonomy, modality, intended use, exclusions	Distinguishes scientific purpose from function name
Inputs	data kind, format, reference space, units, required metadata	Avoids superficial format matching
Outputs	artifact types, coordinate space, units, uncertainty/QC links	Enables correct interpretation
Conditions	preconditions, incompatibilities, required preprocessing	Supports correct selection/deferral
Execution	container digest, command/API, compute resources, permissions	Repeatability and security
Validation	schema checks, test fixtures, QC requirements, unsupported cases	Evidence of tested behavior
Provenance	parameters, versions, input hashes, outputs, execution ID	Reconstruction and auditing

4.4 Example contract (illustrative only)

```
id: neuro.t1.hippocampal_volumes
version: 0.1.0
purpose: Estimate hippocampal volumes from T1w MRI
input:
  kind: structural_mri
  modality: T1w
```

```
  format: nifti
  space: subject_native
preconditions:
  - anatomical_image_available
  - compatible_segmentation_environment
output:
  kind: regional_volume_table
  units: mm3
validation:
  - input_schema_passed
  - output_regions_present
  - qc_report_available
execution:
  approval_required: true
  container: image@sha256:<pinned_digest>
  backend: local_container
```

This illustrative schema is intentionally simple; adapter implementations and anatomical volume extraction must be independently verified before production use.

4.5 Architecture and workflow

1. A user or agent sends a scientific intent query to SciCap discovery.
2. The registry resolves candidate capabilities by task vocabulary, modality, data requirements and software availability.
3. SciCap returns machine-readable contracts, evidence of support and explicit exclusions.
4. The agent selects an operation; deterministic schema/precondition validators inspect the proposed arguments.
5. Permissions are checked, and an authorized executor runs a version-pinned program through an existing engine.
6. Outputs, validation status, provenance and unsupported checks are returned as distinct records.

4.6 Interfaces and repository organization

- packages/scicap-core: schema, type validation and registry search.
- packages/scicap-adapters: MCP, OpenAPI and optional Boutiques/CWL import/export.
- packages/scicap-neuro: maintained neuroimaging capability definitions.
- examples/: synthetic datasets, example invocations and test fixtures.
- benchmarks/: frozen evaluation cases separated from tool runtime code.
- docs/: generated contract reference, install guides, developer tutorials and failure policies.
- website/: read-only search and explorer; avoid user-uploaded PHI.

4.7 Security, risk and operating principles

No arbitrary shell execution from model-produced text. Constrain parameters using allowlisted operations and schemas. Require explicit approval for resource-intensive, destructive or remote execution. Maintain least-privilege backend credentials, data-access checks and resource quotas. Avoid uploading sensitive data to third-party LLM services by default. Document limits: scientific QC not established unless a specified and validated check ran.

5. Datasets, tools, compute and access

5.1 Available external resources

Resource	Use	Acquisition/validation requirement
OpenNeuro [13]	Public BIDS MRI examples and selected real execution tasks	Choose dataset snapshot, capture license, modalities, files and DOI
BIDS examples	Directory/metadata fixtures, quick static tests	Many examples do not include real image content
BIDS validator / PyBIDS	Standard compliance and metadata handling	Pin versions and use unchanged baseline
FreeSurfer / FastSurfer	Anatomical reference processing	Verify licenses, citation, compatible hardware
ANTs / FSL / MRtrix3	Registration and downstream fixtures	Choose minimal test operations; respect license
Boutiques / CWL / Nipype	Descriptor and engine comparators	Use genuine existing features, not artificially weakened baselines
Docker or Apptainer	Execution containerization	Pin image digests and host characteristics

5.2 Experimental dataset tiers

- Tier 0: fully synthetic artifact fixtures and metadata; milliseconds to seconds per task; ideal for unit tests and negative cases.
- Tier 1: public small BIDS datasets for genuine MRI inputs; suitable for independently reproducible integration testing.
- Tier 2: richer public datasets with different scanners/acquisitions; evaluate tool/domain transfer.
- Tier 3 (optional): approved institutional data for external validation; excluded from public replication unless separately permitted.

5.3 Acquisition manifest fields

For every dataset capture dataset ID, snapshot/version, URL/DOI, license, participant count, modalities, acquisition notes, preprocessing, data-access restrictions, expected input artifacts, file hashes, date obtained and required citations. Do not choose data solely because it makes a particular model perform well.

5.4 Current capability assessment

Likely available through prior projects: familiarity with Docker, Python, FastAPI, BIDS conversion, FreeSurfer/FastSurfer, MRtrix and Slurm. Not yet verified: institutional permissions, reusable code ownership, reproducible public reference implementations, licenses, compute allocation, named external collaborators and QC fixture sets. Treat all as audit tasks, not assumed assets.

6. Experimental program: detailed protocol

6.1 Trial unit and study design

Trial unit = one scientific intent, specified data fixture and available tool catalog. Use paired evaluations: identical query, catalog content, agent model, inference settings, compute budget and scoring across intervention conditions. Randomize condition order where feasible. Prevent benchmark leakage into contract wording or prompt tuning. Freeze versioned materials before confirmatory analysis.

6.2 Conditions / baselines

Arm	Available information	What it isolates
A. Documentation	Official manuals / readmes + permitted commands	Human-oriented conventional source material
B. Standard tools	MCP/OpenAPI description and ordinary JSON input schema	Existing programmatic agent accessibility
C. Rich existing descriptors	Boutiques/CWL/Nipype-native typed interfaces when appropriate	Competes with mature description standards
D. SciCap contract	Same execution tools plus scientific semantic fields	Incremental value of SciCap representation
E. Ablations	D with selected contract fields removed	Causal contribution of individual metadata groups

Baseline equivalence is crucial: adapter quality, available tools, underlying executors and effective information budgets should be documented and as comparable as possible.

6.3 Task families and suggested pilot allocations

Category	Pilot tasks	Reference evaluation
Tool discovery	10	Correct operation and rationale with permitted alternatives
Parameter configuration	10	Validated argument sets; required constraints met
Precondition identification	10	Recognizes missing inputs and required preprocessing
Correct abstention/clarification	10	Declines or flags unsupported/unavailable operation
End-to-end execution	10	Correct artifact plus reproducibility/QC record

Start with approximately 50 pilot tasks; expand only after inspecting failure patterns and estimating uncertainty. A possible confirmatory phase is 120–240 independently annotated tasks with multiple runs per condition and stratification by difficulty. These are planning ranges, not a sample-size justification.

6.4 Reference truth and expert review

- Each task gets a machine-verifiable expected operation, admissible parameter set, prerequisites, known invalid choices and correct deferral behavior.
- At least two qualified reviewers independently label a representative subset; adjudicate disagreements and report agreement.
- Separate objective machine scoring from subjective reviewer judgment.
- Use reference implementations and validated fixtures to test produced artifacts.
- Maintain sealed held-out tasks and tools never shown in developer prompts, unit tests, or example docs.

6.5 Cross-domain stretch experiment

To support broad computational-science journals, demonstrate capability contracts in a domain outside neuroimaging (e.g., bioinformatics sequence QC, geospatial raster processing or materials simulation), selected with an expert collaborator. Reuse the identical core schema and scoring framework; document any field or ontology modifications. If this experiment is unavailable, use a neuroimaging journal or software venue rather than claiming generality.

6.6 Repetitions and model choice

Evaluate at least two distinct model families where practical; use temperature/determinism settings reported precisely. Repeat stochastic trials with paired seeds where supported, noting providers may not guarantee strict reproducibility. Preserve model/version IDs, prompts, tools, outputs, cost and timestamps. Ensure model comparisons do not confound context window or hidden tools.

7. Metrics, statistical plan and preregistration

7.1 Primary metric

Recommended primary endpoint: proportion of held-out tasks that produce a scientifically correct end-state (correct selection and valid arguments, or justified abstention), as defined by adjudicated task-specific criteria. Distinguish “process completed” from “scientific task correctly completed.”

7.2 Secondary outcomes

Outcome	Operational definition
Selection accuracy	Chosen tool is among admissible reference capabilities
Argument validity	All required inputs, parameter constraints and type checks pass
Precondition recognition	Correctly identifies missing prerequisites
Unsafe/invalid attempt rate	Attempts prohibited or scientifically incompatible invocation
Appropriate deferral	Abstains when capability/data/permission is insufficient
Execution success	Reference output generated and checks passed
Reproducibility completeness	Proportion of required provenance fields captured
Efficiency	Tokens, API cost, wall time, tool calls, compute time
Authoring burden	Time to publish and maintain each contract; defect frequency

7.3 Statistical analysis plan

- Use paired confidence intervals for differences in task success rates, with task-level resampling or appropriate mixed-effects modeling to account for repeated trials, models and task families.
- Report effect sizes and interval estimates rather than significance thresholds alone; if hypothesis tests are used, declare primary comparison and multiplicity handling.
- Calculate pilot-based precision or power requirements using expected paired disagreement; do not treat every repeated model run as an independent new task.
- Analyze task-category heterogeneity, complexity, catalog size and held-out tool performance.
- Report calibration of abstention, false acceptance of invalid requests, and failure severity.
- Keep exploratory analyses explicitly exploratory; publish negative findings and failure examples.

7.4 Ablation grid

Ablation	Purpose
No semantic-purpose text	Tests whether intent vocabulary contributes beyond function names
No modality/type constraints	Tests value of domain input semantics
No scientific preconditions	Tests missing-prerequisite recognition
No output semantics	Tests downstream interpretation support
No example calls	Separates schema value from demonstration value
Plain-language descriptions matched for length	Tests representation vs extra token volume

7.5 Threats to validity

- Information inequality: SciCap condition may convey more words than baseline; counter with length-matched and human-readable semantic baseline.
- Evaluation leakage: tasks adapted from tool examples; use held-out authors and task construction.

- Synthetic-task bias: easy fixtures overestimate real-world applicability; validate with genuine MRI executions.
- Adapter artifacts: substandard comparison adapters unfairly penalize baselines; conduct implementation review.
- Expert circularity: schema authors also label task correctness; use independent raters and blinded outputs.
- Hardware variability: control/pin environment and report execution variance.
- Provenance and version drift: archive immutable reference packages and model response logs subject to terms.

8. Tool implementation, community and public release

8.1 First-release milestones

Release	Features	Exit criteria
v0.1 alpha	Schema, CLI, 5 contracts, synthetic fixtures	Unit/schema tests and install guide pass on clean machine
v0.2 research preview	MCP adapter, public catalog, dry run, execution sandbox	External researcher can reproduce 3 example workflows
v0.3 beta	Expanded catalog, provenance, import adapters, error taxonomy	Security review and stable integration suite
v1.0	Versioned schema, upgrade policy, docs and governance	Documented use, CI and maintained releases

8.2 Quality and governance

- License selection after institution/IP review; record all third-party code and data licenses.
- CI should validate schemas, evaluate examples, run tests, check formatting and prevent accidental secrets inclusion.
- Use semver for package plus independent version identifiers for schema and capability manifests.
- Add contributor guide, issue template, code of conduct, security contact, changelog, citation metadata and maintainer responsibilities.
- Use DOI archival (e.g., Zenodo) for exact paper-linked software release; never rely on a moving main branch.
- Provide consent-based feedback and avoid PHI uploads or undisclosed telemetry.

8.3 Adoption evaluation without conflating it with validity

Log opt-in, aggregate uptake where permitted: independent installations, external contracts, validated integration pull requests, repeat users and documentation issues. These demonstrate utility, not scientific correctness. Keep public user feedback separate from randomized confirmatory evaluation. Release the methodology even if performance hypotheses fail.

8.4 Example demo journey

7. Researcher browses “T1 hippocampal volume estimation” in catalog.
8. SciCap shows required data, preprocessing, expected volumes and QC obligations.
9. Researcher exports a contract and validates an example input locally.
10. Agent requests execution, but sees missing-required-input or QC status if applicable.
11. After explicit permission, local tool runs and emits versioned provenance plus an output manifest.

9. Manuscript construction plan

9.1 Recommended working title and abstract shell

Title: Scientific Capability Contracts Improve [measured outcome] in AI-Assisted Computational Workflows. Do not include “improve” until confirmed results establish it; the neutral preregistration title is “Evaluating Scientific Capability Contracts for AI-Agent Discovery and Execution.”

Abstract structure: (i) fragmented scientific software and machine-actionability gap; (ii) proposed contract and open SciCap implementation; (iii) controlled comparison against documentation, conventional schemas and existing descriptors; (iv) primary performance and generalization results with effect estimates; (v) limitations and reproducibility artifacts. No numerical results should be written until experiments are complete.

9.2 Full manuscript outline

- Introduction — Complex scientific tools; human documentation vs agent tools; limits of syntax-only schema; precise gap; 2–3 research claims.
- Related work — Workflow systems, neuroimaging pipelines, BIDS, Boutiques, MCP, semantic web services, agentic workflows and science agents; fair prior-art table.
- Methods: capability model — Formal definitions, schema fields, semantics, versioning, examples, mapping to baseline standards and validation boundaries.
- Methods: reference implementation — Architecture, registry, adapters, executor isolation, security, reproducibility and public releases.
- Methods: benchmark — Task sampling, datasets, expert annotations, model conditions, prompts, baseline adapters, randomization, frozen splits.
- Methods: statistics — Primary endpoint, paired design, intervals, power/precision, repeated measures and exclusions.
- Results 1: discovery — Selection and configuration accuracy with task-category breakdown and confidence intervals.
- Results 2: ablations — Which semantic fields matter and whether token-matched prose suffices.
- Results 3: transfer — Held-out tools, data and cross-domain application where available.
- Results 4: costs/failures — Compute, API cost, catalog authoring, security/abstention and failure taxonomy.
- Discussion — What was learned, why systems benefited or not, boundaries, compatibility with existing standards.
- Limitations / ethics — Expert annotations, generality, security, clinical restrictions, adoption caveats and reproducibility.
- Data and code availability — Persistent versioned repositories and permissions; model logs if license permits.

9.3 Figures and tables

Display	Purpose	Minimum content
Figure 1	Architecture	Intent → registry → schema validation → authorized executor → provenance
Figure 2	Scientific capability contract	Fields, semantically meaningful example and existing descriptor mapping
Figure 3	Primary performance	Paired success-rate estimates + uncertainty across baselines
Figure 4	Ablation	Impact of each scientific-contract field
Figure 5	Generalization	Unseen tools, modalities, datasets and domain transfer
Figure 6 / Table	Failure/efficiency	Abstention, errors, runtime, tokens, contract authoring

Supplementary	Exact reproducibility	Task prompts, labels, all configs and per-task outputs

9.4 Supplementary material

- Schema JSON, design decisions and compatibility mapping to Boutiques/CWL/MCP.
- Complete task taxonomy, selection rules, frozen task IDs and scoring rubric.
- Dataset snapshot manifest with DOI, licenses, checksums and exclusions.
- Exact prompts, model settings and deployment hashes subject to license.
- Full baseline descriptions; independent reviewer agreement and adjudication.
- Complete performance tables, negative examples, compute accounting and additional sensitivity analyses.
- Container recipe, test scripts, instructions for reproducing key figures.

9.5 Manuscript writing order

Write Methods and related-work matrix before major experiments; preregister endpoints and scoring; record software decisions as they happen; write Results from final analysis scripts; only then finalize title, abstract, significance and journal cover letter. Avoid post-hoc selective reporting.

10. Journal strategy: highest defensible tier

Tier / outlet	What must be demonstrated	Practical positioning
Stretch: Nature Computational Science	Novel computational framework with substantive methodology, cross-domain generalization, rigorous baselines, impactful science applications, reproducible evidence	Target only if breadth and effect are convincing; scope is explicitly cross-disciplinary [14]
High: Nature Communications	Important advance for specialists, strong controlled study, compelling external validation, independent reproduction	Appropriate if significant new agent-science systems insight [15]
Specialist high: Imaging Neuroscience / NeuroImage	Methodologically valuable neuroimaging advances plus realistic pipeline and data validation	Best if study remains primarily neuroimaging
Systems/software: GigaScience / SoftwareX	Well-supported reusable framework with substantial scientific utility and transparent software engineering	Good for strong software contribution with limited method novelty
Software: JOSS	Mature open-source package, research impact, docs/tests, at least six months sustained public development [16]	Not an original experiment venue; publish a distinct software paper

10.1 Nature Computational Science submission pathway

Journal scope emphasizes development and cross-disciplinary applications of computational techniques. The current Article content guidance states up to 3,500 main-text words excluding abstract, Methods, references and legends, a 150-word abstract and up to six main display items; verify guidance again before submission [14,17]. Requirements include replication-capable Methods, author/accountability disclosures, detailed references and a cover letter explaining relevance to diverse readership [18].

10.2 Submission gating decision

- Stretch: cross-domain validation, clear method new to the literature, large externally reviewed benchmark, substantially better outcomes than existing descriptor baseline, compelling use case.

- High specialist: robust improvements in neuroimaging and independent MRI datasets, but limited cross-domain generalization.
- Software outlet: practical SciCap adopted or reused externally, but no compelling demonstrated gain over established methods.
- No submission yet: only working demo or benchmark biased toward proposed system; continue validation.

11. Review risks and responses

Likely reviewer criticism	Concrete response in study
“Boutiques already does this.”	Map field-by-field; compare with genuine Boutiques descriptors; show added semantic benefit and avoid wrapper claims.
“MCP already provides agent tool discovery.”	Use MCP as common transport in B and D, vary only scientific metadata.
“You fed your method more information.”	Equal information budget controls and length-matched plain-text semantic descriptions.
“Agent performance depends on one model.”	Multiple model families and disclosed model-specific effects.
“The tasks are too easy or cherry-picked.”	Expert-reviewed stratified tasks, held-out tools and prespecified selection.
“Task execution is not scientifically valid.”	Explicit scientific truth criteria and artifact checks; honest unknown/QC-not-run statuses.
“This works only in your lab.”	Public reproductions, clean-room installation and cross-site/domain test where feasible.
“You built a new engine unnecessarily.”	Reuse well-supported existing executors; contract is separate from compute orchestration.
“The release cannot be maintained.”	Governance, package tests, compatibility policy, contribution pathway and external trial.

12. Milestones, dependencies, ownership and costs

12.1 Milestone plan

Period	Engineering	Research/paper gate
Weeks 1–2	Inventory tools and ownership; choose MVP capabilities	Expanded prior-art matrix; lock intended contribution
Weeks 3–4	Schema v0.1 and offline catalog with tests	Expert task taxonomy and outcome rubric
Weeks 5–6	MCP/Boutiques/CWL adapter examples	Pilot data fixtures and baseline equivalence review
Weeks 7–9	Local executor, provenance, security boundary	50-task pilot, issue analysis and annotation check
Weeks 10–12	Docs/demo and first public alpha	Precision/power plan, preregister confirmatory protocol
Months 4–6	Expand contracts and independent integrations	Frozen controlled experiment, held-out tasks, analysis
Months 7–9	Stable release, outside-user feedback	Manuscript, internal replication, submit to suitable tier

12.2 Minimum collaborators

- Research engineering lead: schemas, implementation, benchmarks, experiments, results provenance and manuscript.
- Neuroimaging scientist: correctness criteria, QC assumptions, reference workflow review and independent task adjudication.
- Scientific workflow/semantic systems specialist: prior-art comparison and technically defensible abstraction.
- Statistics/evaluation collaborator: matched evaluation design, effect estimation, multiplicity and reproducibility.
- Optional cross-domain expert: supports Nature Computational Science ambition with independent use case.

12.3 Budget categories to estimate before experiments

Cost	Sizing method
Agent API or local inference	N tasks × arms × models × repeats × measured token/unit cost
CPU/HPC	Reference workflow CPU-hours per task × execution repetitions
Storage	MRI data, frozen outputs, containers, logs, model traces
Annotation	Number of tasks × expert review time × rates/availability
Engineering time	Schema/adapter hours, security review, docs and ongoing maintenance
Publishing and hosting	Journal fees if applicable, archival, domain, cloud demonstrations

No dollar estimates are stated because exact model, compute, staffing and journal options are not yet fixed. Do not confuse an illustrative task count with power analysis.

12.4 Permissions and institutional separation

Document which existing NeuroInsight components were created under university employment, how work time and HPC are funded, whether grant terms apply, and which datasets/code may be redistributed. Obtain authorized determinations before importing restricted code into a public product or commercial organization. Public synthetic and OpenNeuro datasets should be the default research distribution where licenses permit.

13. Literature search protocol (to complete systematic novelty review)

13.1 Search sources and strings

Sources: PubMed, Google Scholar, Semantic Scholar/OpenAlex, ACM Digital Library, IEEE Xplore, arXiv, bioRxiv, Nature portfolio, workflow standards and official software documentation. Search in English; deduplicate by DOI and title, keep version-of-record preference, and record retrieval dates.

- ("scientific workflow" OR "research workflow") AND ("semantic" OR "ontology" OR "type system") AND (validation OR composition)
- ("LLM agent" OR "AI agent") AND ("tool discovery" OR "tool selection") AND (benchmark OR metadata)
- (MCP OR OpenAPI OR Boutiques OR CWL OR Nipype) AND ("scientific" OR "neuroimaging") AND (descriptor OR reproducibility)
- ("science automation" OR "agentic science") AND (workflow generation OR execution OR verification)
- (BIDS OR neuroimaging) AND (workflow interoperability OR semantic constraints OR agent)

13.2 Inclusion and extraction process

12. Collect foundational standards and representative recent papers, then snowball backward and forward from the closest 10–15 works.
13. Record year, venue, DOI/URL, capability encoding, domain, agent model, execution engine, validity checks, data, metrics and code availability.
14. Classify each work: interfaces, semantic discovery, workflow composition, execution, scientific validation or agent benchmarking.
15. Independently verify potential novelty blockers and update the claim matrix.
16. Define a literature search cutoff at manuscript submission and include relevant intervening 2026–2027 papers.

13.3 Critical novelty check

The decisive review must cover Boutiques and semantic service descriptors, not just generic agent papers. If an existing standard already supports every proposed contract field, novelty shifts to controlled scientific-agent outcomes, usability or domain-specific validation—not a new schema itself.

14. Annotated reading list and verification links

[1] Gorgolewski et al. (2011). Nipype: A flexible, lightweight and extensible neuroimaging data processing framework in Python. Frontiers in Neuroinformatics. DOI: 10.3389/fninf.2011.00013.

Contribution to this project: Core prior work on typed tool interfaces, workflow graphs, execution.

https://doi.org/10.3389/fninf.2011.00013

[2] Gorgolewski et al. (2016). The Brain Imaging Data Structure. Scientific Data. DOI: 10.1038/sdata.2016.44.

Contribution to this project: Neuroimaging data organization and machine-readable metadata.

https://doi.org/10.1038/sdata.2016.44

[3] Gorgolewski et al. (2017). BIDS Apps. PLOS Computational Biology. DOI: 10.1371/journal.pcbi.1005209.

Contribution to this project: Standard neuroimaging executable interface and reproducibility.

https://doi.org/10.1371/journal.pcbi.1005209

[4] Glatard et al. (2018). Boutiques: A flexible framework to integrate command-line applications in computing platforms. GigaScience. DOI: 10.1093/gigascience/giy016.

Contribution to this project: Most essential counterexample to simplistic “new descriptor” claims.

https://doi.org/10.1093/gigascience/giy016

[5] Esteban et al. (2019). fMRIPrep: a robust preprocessing pipeline for functional MRI. Nature Methods. DOI: 10.1038/s41592-018-0235-4.

Contribution to this project: Model for mature automated pipelines and validation.

https://doi.org/10.1038/s41592-018-0235-4

[6] Common Workflow Language. User Guide / specification.

Contribution to this project: Existing tool/workflow machine-readable contracts; evaluate import mapping.

https://www.commonwl.org/user_guide/

[7] RO-Crate. Research Object Crate specification.

Contribution to this project: Packaging research metadata and scientific provenance.

https://www.researchobject.org/ro-crate/specification/

[8] W3C PROV-O. Provenance Ontology.

Contribution to this project: Formal vocabulary for agents, activities, artifacts, derivations.

https://www.w3.org/TR/prov-o/

[9] Model Context Protocol. Tools specification (2025-06-18 reference).

Contribution to this project: Tool discovery/execution transport baseline; use current specification for implementation.

https://modelcontextprotocol.io/specification/2025-06-18/server/tools

[10] OpenAPI Initiative. OpenAPI specification.

Contribution to this project: Standard structured programmatic service descriptions.

https://spec.openapis.org/oas/latest.html

[11] Balis et al. (2026). From Research Question to Scientific Workflow: Leveraging Agentic AI for Science Automation. arXiv:2604.21910.

Contribution to this project: Very close prior work on LLM intent, domain skills and deterministic workflow generation.

https://arxiv.org/abs/2604.21910

[12] Miao et al. (2026). Reimagining research papers as interactive and reliable AI agents. Nature. DOI: 10.1038/s41586-026-11044-y.

Contribution to this project: Contemporary paper-to-agent MCP conversion and evaluation; adjacent not identical.

https://doi.org/10.1038/s41586-026-11044-y

[13] OpenNeuro public dataset user guide.

Contribution to this project: Public BIDS data source with snapshots and citation guidance.

https://docs.openneuro.org/user-guide/

[14] Nature Computational Science. Aims and scope.

Contribution to this project: Scope evidence for stretch journal.

https://www.nature.com/natcomputsci/aims

[15] Nature Communications. Aims and scope.

Contribution to this project: Specialist significance target and multidisciplinary reach.

https://www.nature.com/ncomms/aims

[16] Journal of Open Source Software. Submission and review criteria.

Contribution to this project: Public development duration, software maturity and research use.

https://joss.readthedocs.io/en/latest/submitting.html

[17] Nature Computational Science. Article content types.

Contribution to this project: As-of-2026 Article format; recheck before submitting.

https://www.nature.com/natcomputsci/content

[18] Nature Computational Science. Submission preparation requirements.

Contribution to this project: Replication materials, cover letter, AI-use reporting and manuscript requirements.

https://www.nature.com/natcomputsci/submission-guidelines

[19] WorkflowHub. Registry.

Contribution to this project: Existing workflow discovery and reuse ecosystem.

https://workflowhub.org/

[20] Nipoppy. Framework and documentation.

Contribution to this project: Related neuroimaging data and pipeline organization framework.

https://nipoppy.readthedocs.io/

15. Submission-ready completion checklist

Novelty

- ☐ Prior-art matrix updated through submission cutoff
- ☐ Boutiques/CWL/Nipype equivalence reviewed
- ☐ One falsifiable claim and scope of generality fixed

Software

- ☐ Core package reproducibly installs
- ☐ Manifest specification and test suite versioned
- ☐ Five+ public capabilities work with fixtures
- ☐ MCP adapter tested under approvals
- ☐ Provenance, safety and error reporting documented

Experiments

- ☐ Frozen task split and expert ground truth
- ☐ Baselines fairly implemented and documented
- ☐ Pilot-based sample-size/precision rationale
- ☐ Matched multi-model evaluation completed
- ☐ Held-out tool test and field ablation completed
- ☐ Failures and negative findings reported

Dissemination

- ☐ Independent replication completed
- ☐ Public data/code release rights confirmed
- ☐ Readme/tutorials/citation/license/change log complete
- ☐ Figures and manuscript generated from frozen analysis scripts
- ☐ Journal checklist, AI disclosure and cover letter complete

Bottom-line decision

Start with a tested SciCap capability contract and adapter library, using existing standards as dependencies. Conduct a controlled evaluation that asks whether domain-specific scientific contracts yield better end-state decisions than mature tool interfaces. The publication tier should emerge from effect size, comparative novelty, transfer across tools/domains, and independent reproducibility—not from the sophistication of the demo or the journal aspiration alone.

Bibliography links were checked selectively on 25 October 2026; the literature list is curated, not exhaustive. Journal policies can change and must be rechecked on the day of submission.